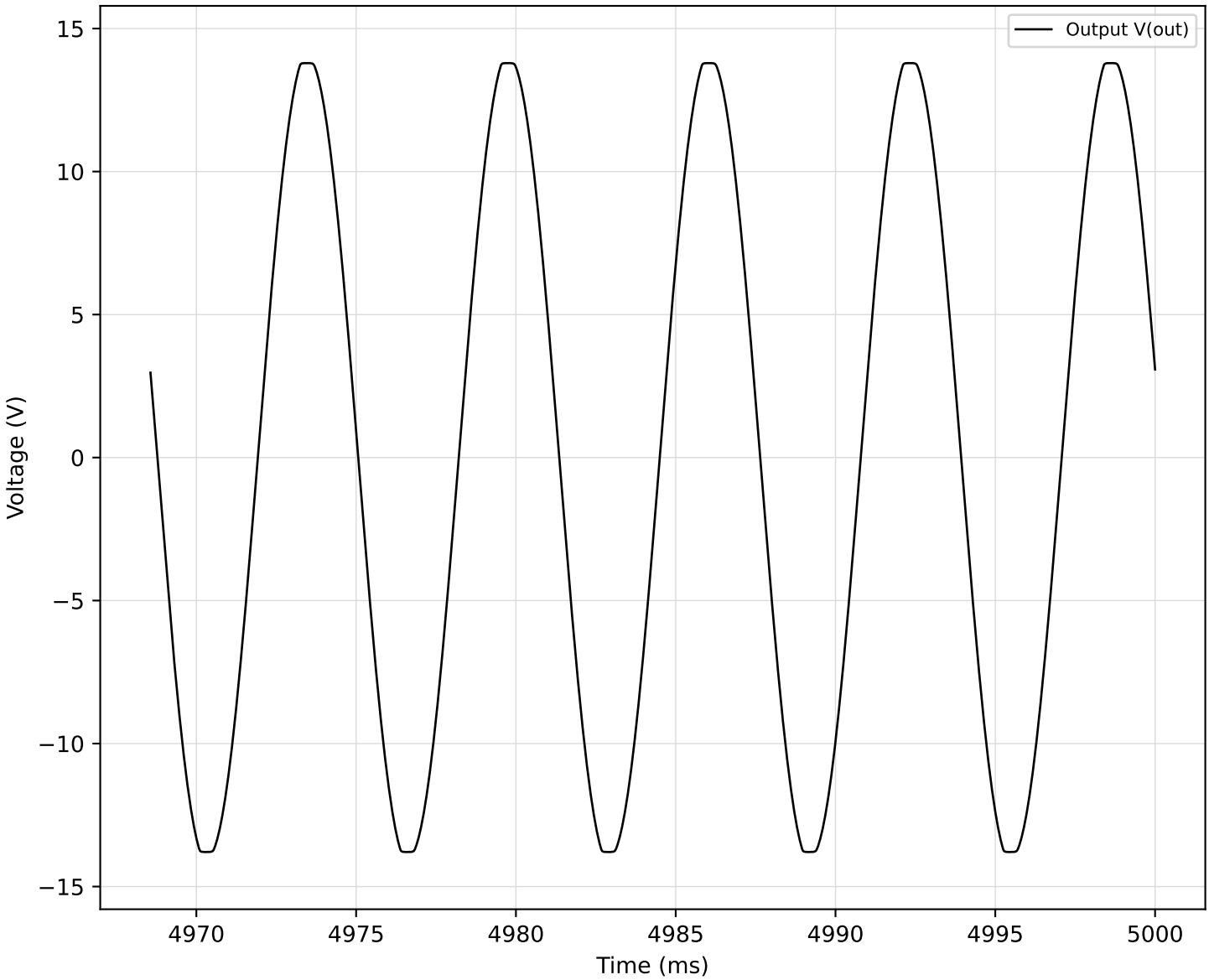
EC3091D - Experiment 8

Wien - Trial 1 / Figure 1

R = 10000 ohm; C = 0.1 uF; supplies = +/-15 V

NATIONAL SEMICONDUCTOR LM741 MACROMODEL



T = 6.2887 ms; f = 159.015 Hz; Vpp = 27.586 V

Measured from ngspice numerical transient data; 0 V output crossings.

Data: simulation_data/wien_01.txt

Observation / handwritten notes: _____

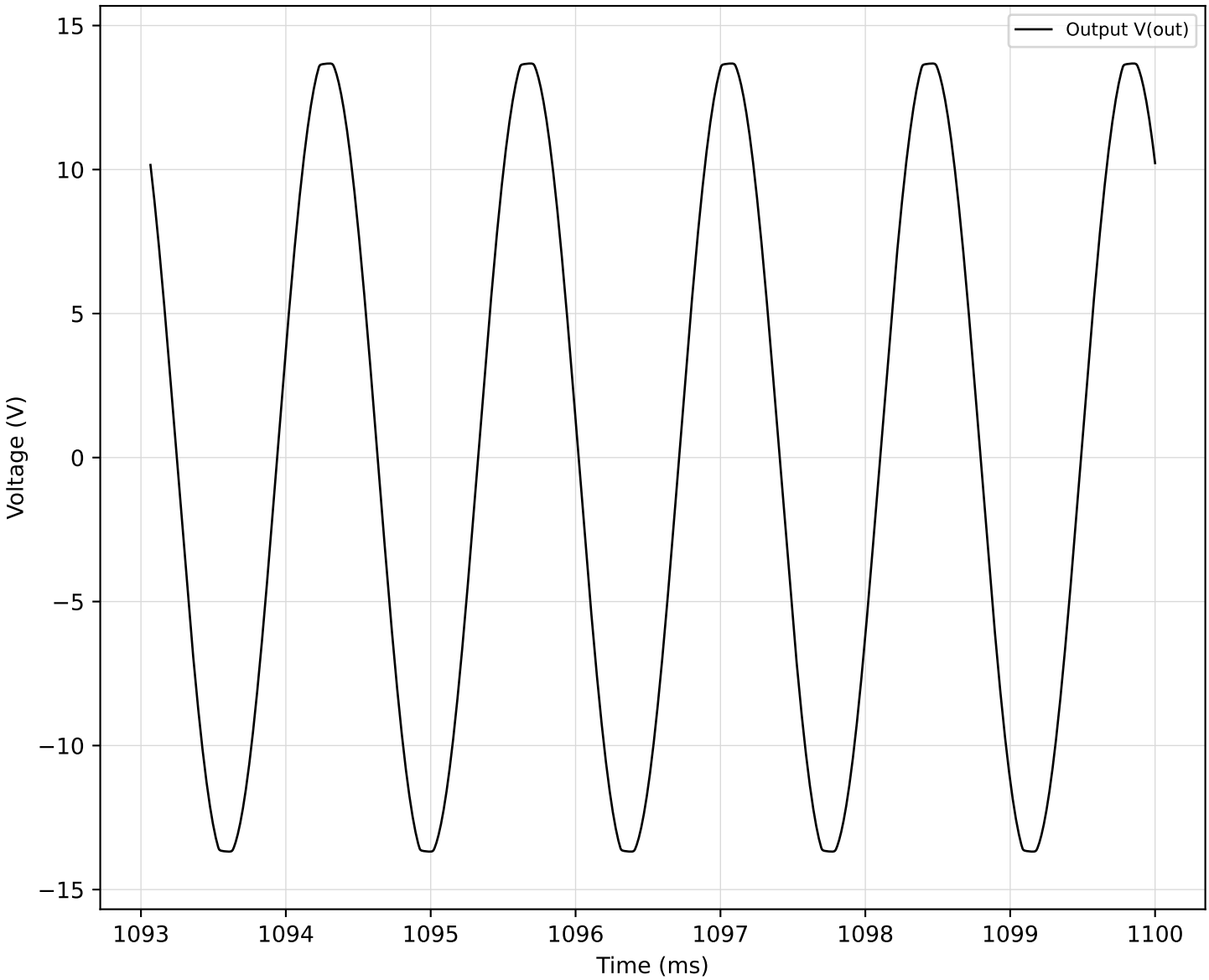
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EC3091D - Experiment 8

Wien - Trial 2 / Figure 1

R = 1000 ohm; C = 0.22 uF; supplies = +/-15 V

NATIONAL SEMICONDUCTOR LM741 MACROMODEL



T = 1.3872 ms; f = 720.890 Hz; Vpp = 27.367 V

Measured from ngspice numerical transient data; 0 V output crossings.

Data: simulation_data/wien_02.txt

Observation / handwritten notes: _____

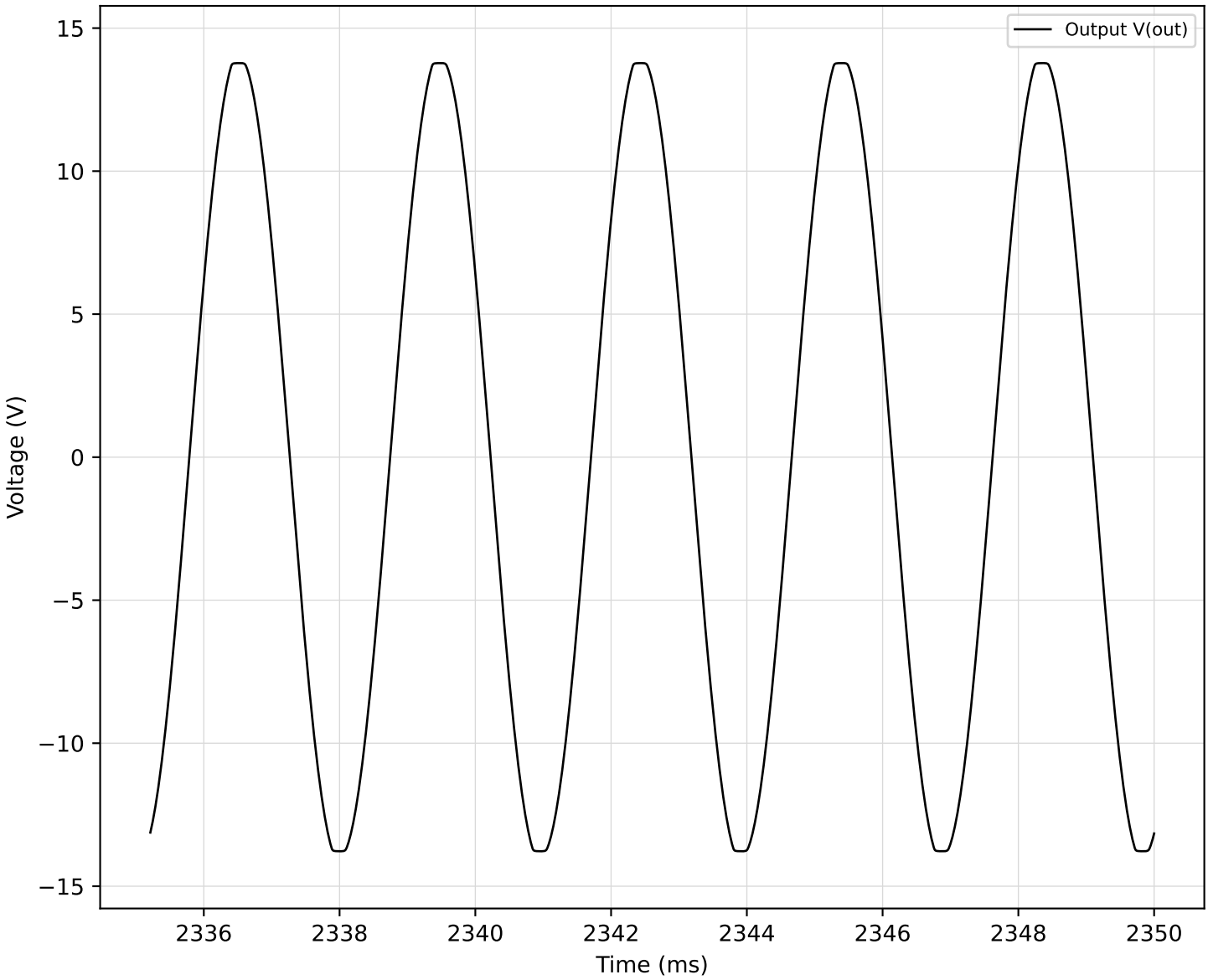
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EC3091D - Experiment 8

Wien - Trial 3 / Figure 1

R = 4700 ohm; C = 0.1 uF; supplies = +/-15 V

NATIONAL SEMICONDUCTOR LM741 MACROMODEL



T = 2.9582 ms; f = 338.049 Hz; Vpp = 27.556 V

Measured from ngspice numerical transient data; 0 V output crossings.

Data: simulation_data/wien_03.txt

Observation / handwritten notes: _____

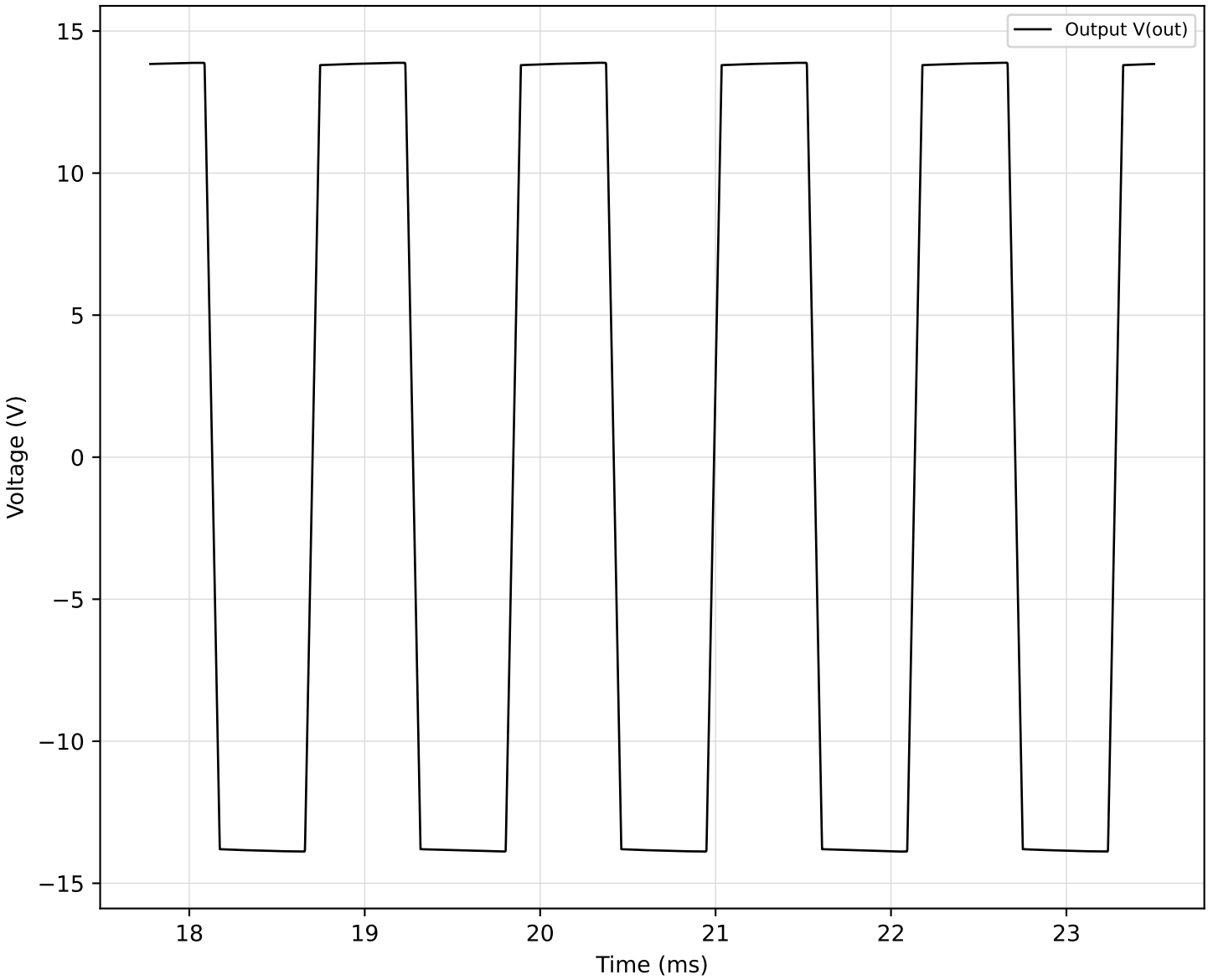
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EC3091D - Experiment 8

Astable - Trial 1 / Figure 2

R = 4700 ohm; C = 0.1 uF; supplies = +/-15 V

NATIONAL SEMICONDUCTOR LM741 MACROMODEL



T = 1.1444 ms; f = 873.788 Hz; Vpp = 27.774 V

Measured from ngspice numerical transient data; 0 V output crossings.

Data: simulation_data/astable_01.txt

Observation / handwritten notes: _____

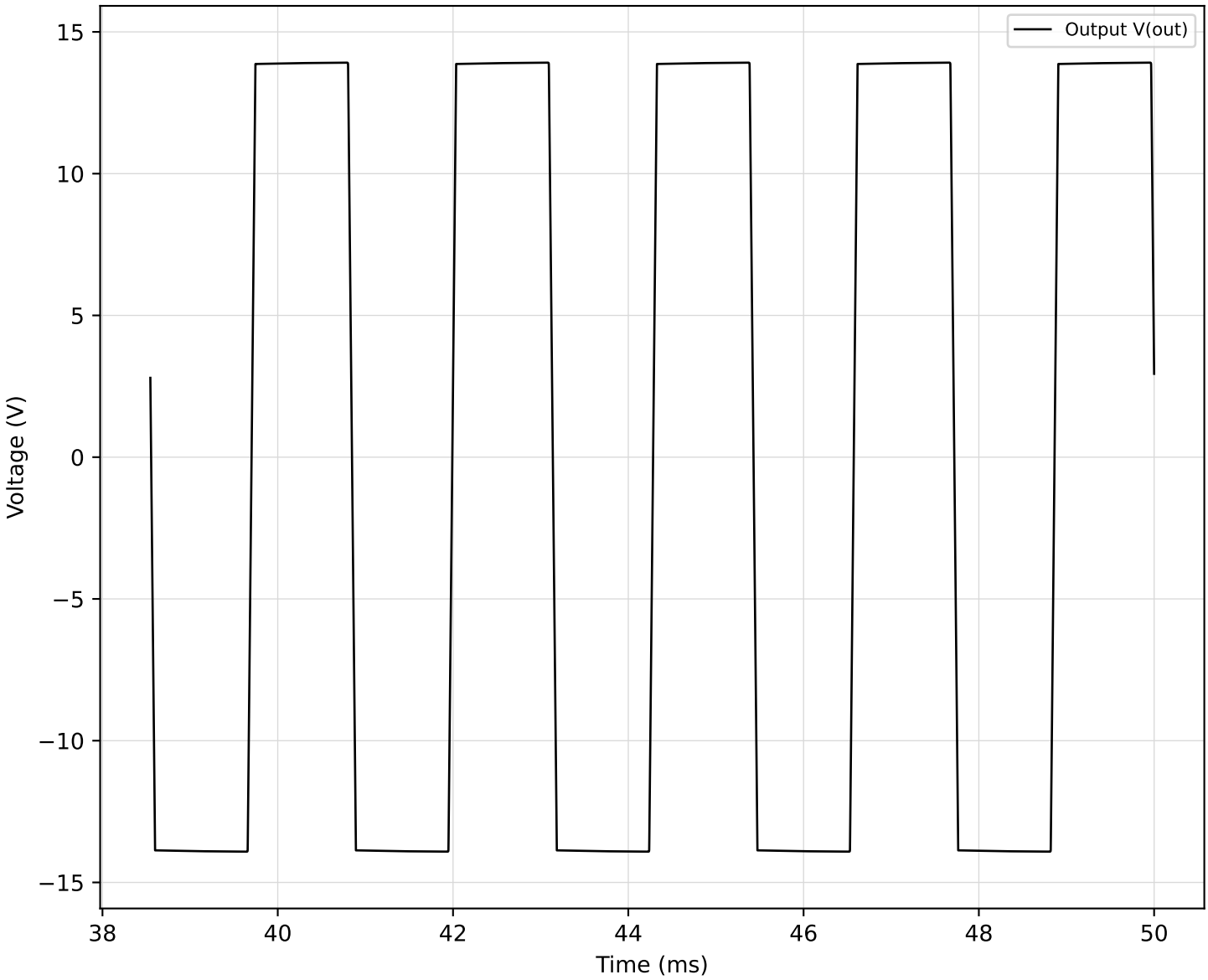
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EC3091D - Experiment 8

Astable - Trial 2 / Figure 2

R = 10000 ohm; C = 0.1 uF; supplies = +/-15 V

NATIONAL SEMICONDUCTOR LM741 MACROMODEL



T = 2.2908 ms; f = 436.519 Hz; Vpp = 27.837 V

Measured from ngspice numerical transient data; 0 V output crossings.

Data: simulation_data/astable_02.txt

Observation / handwritten notes: _____

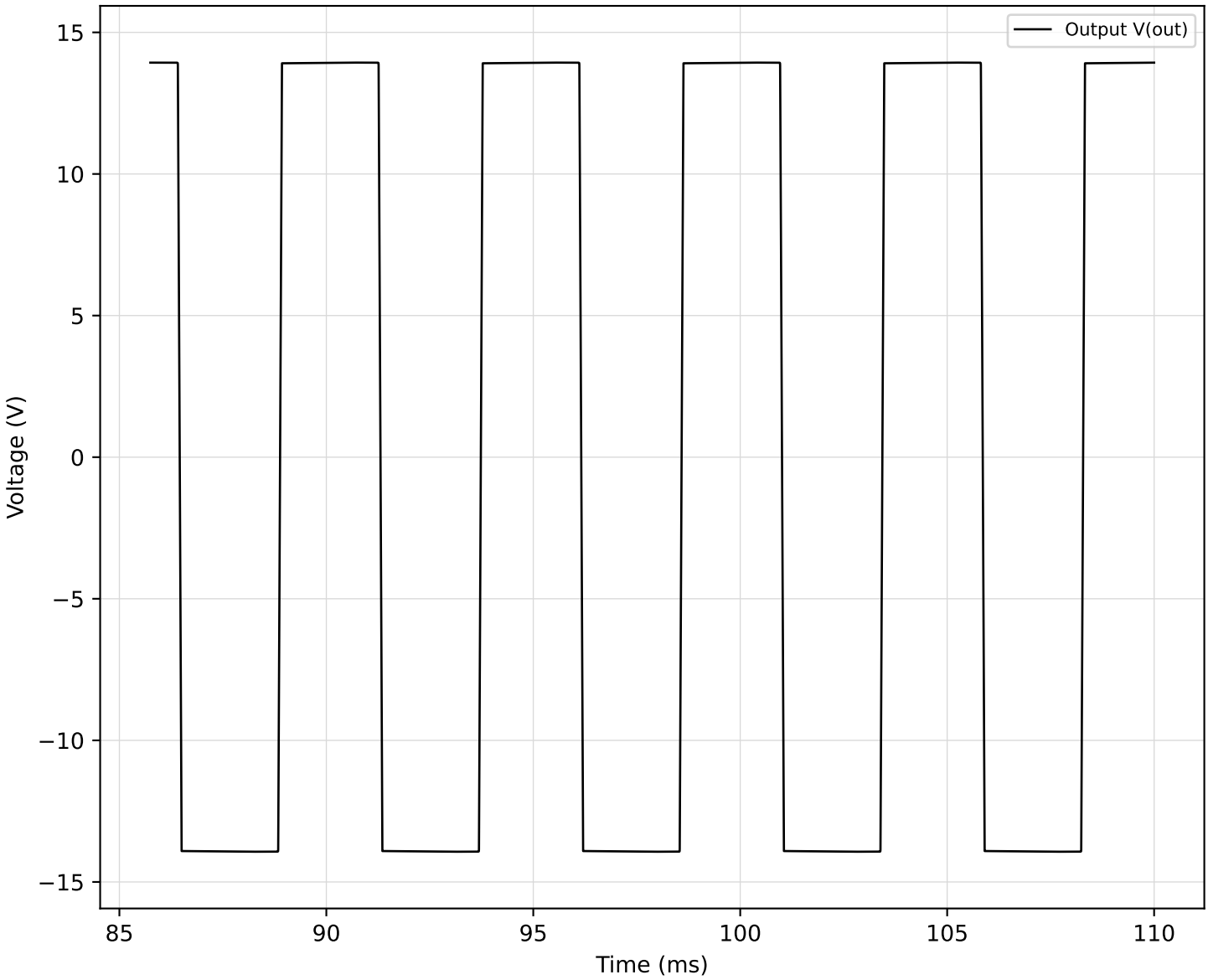
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EC3091D - Experiment 8

Astable - Trial 3 / Figure 2

R = 22000 ohm; C = 0.1 uF; supplies = +/-15 V

NATIONAL SEMICONDUCTOR LM741 MACROMODEL



T = 4.8505 ms; f = 206.166 Hz; Vpp = 27.872 V

Measured from ngspice numerical transient data; 0 V output crossings.

Data: simulation_data/astable_03.txt

Observation / handwritten notes: _____

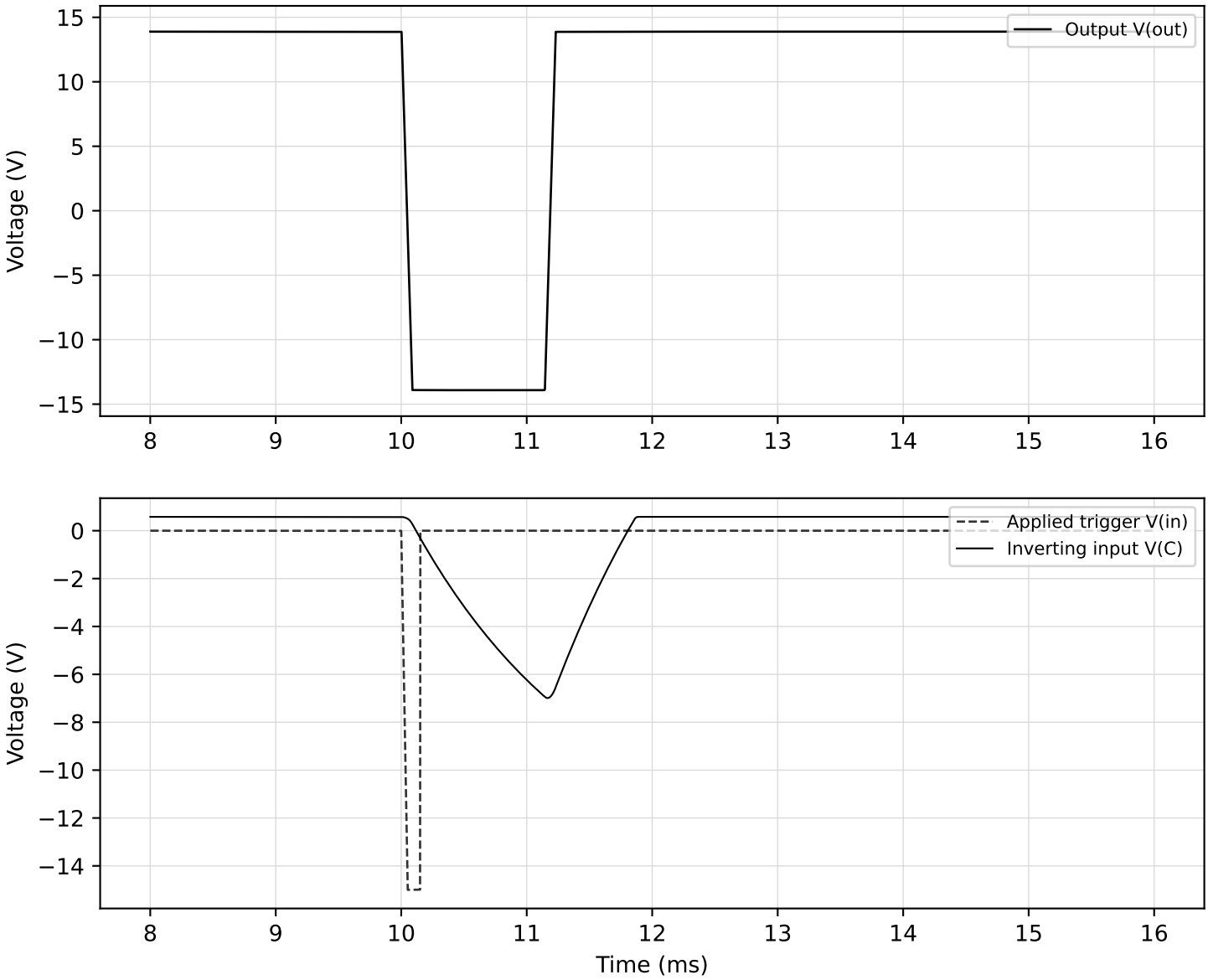
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EC3091D - Experiment 8

Monostable - Trial 1 / Figure 3

R = 15000 ohm; C = 0.1 uF; supplies = +/-15 V

NATIONAL SEMICONDUCTOR LM741 MACROMODEL



LOW pulse = 1.1422 ms; trigger = 50 Hz (external)

Measured from ngspice numerical transient data; 0 V output crossings.

Data: simulation_data/monostable_01.txt

Observation / handwritten notes: _____

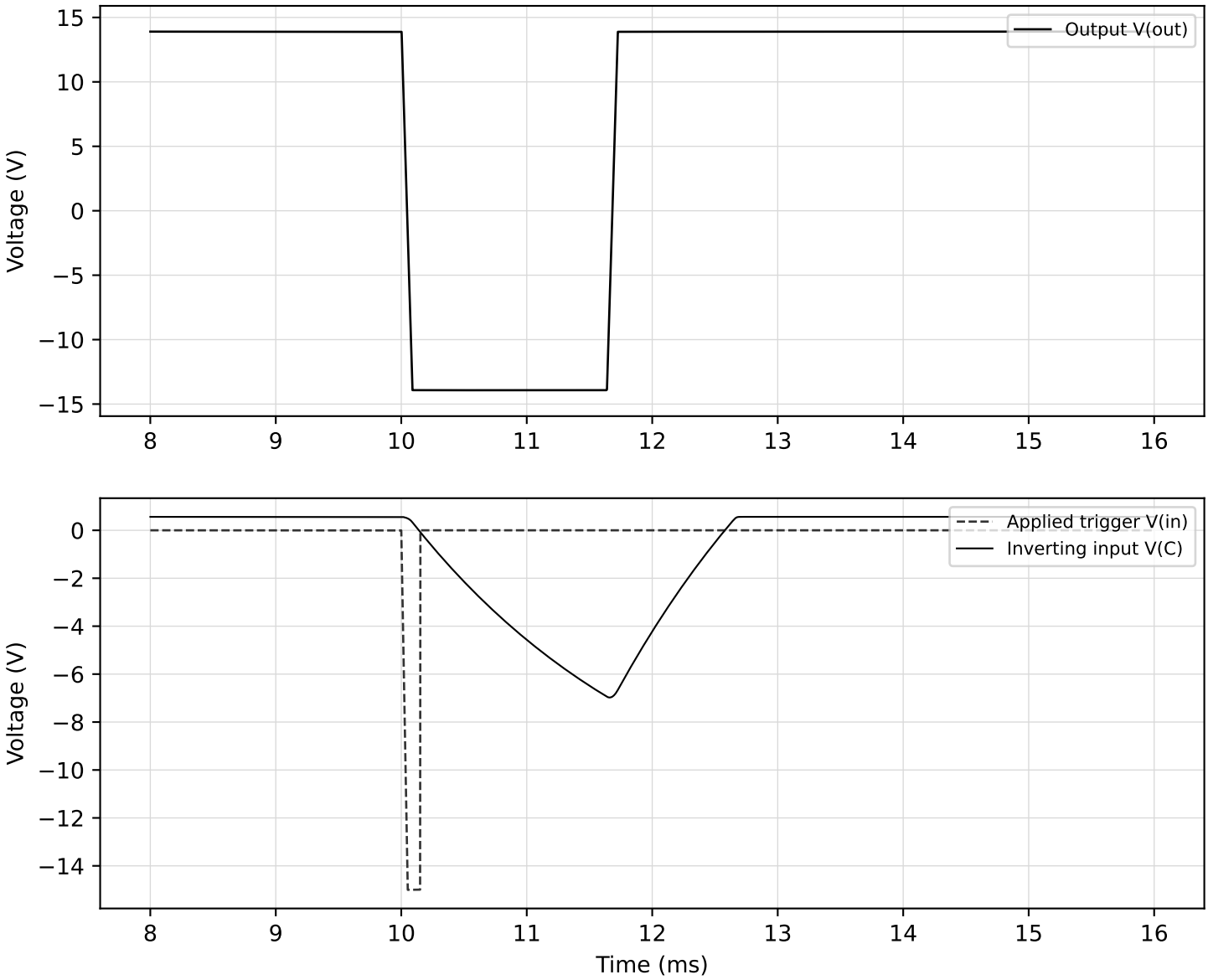
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EC3091D - Experiment 8

Monostable - Trial 2 / Figure 3

R = 22000 ohm; C = 0.1 uF; supplies = +/-15 V

NATIONAL SEMICONDUCTOR LM741 MACROMODEL



LOW pulse = 1.6365 ms; trigger = 50 Hz (external)

Measured from ngspice numerical transient data; 0 V output crossings.

Data: simulation_data/monostable_02.txt

Observation / handwritten notes: _____

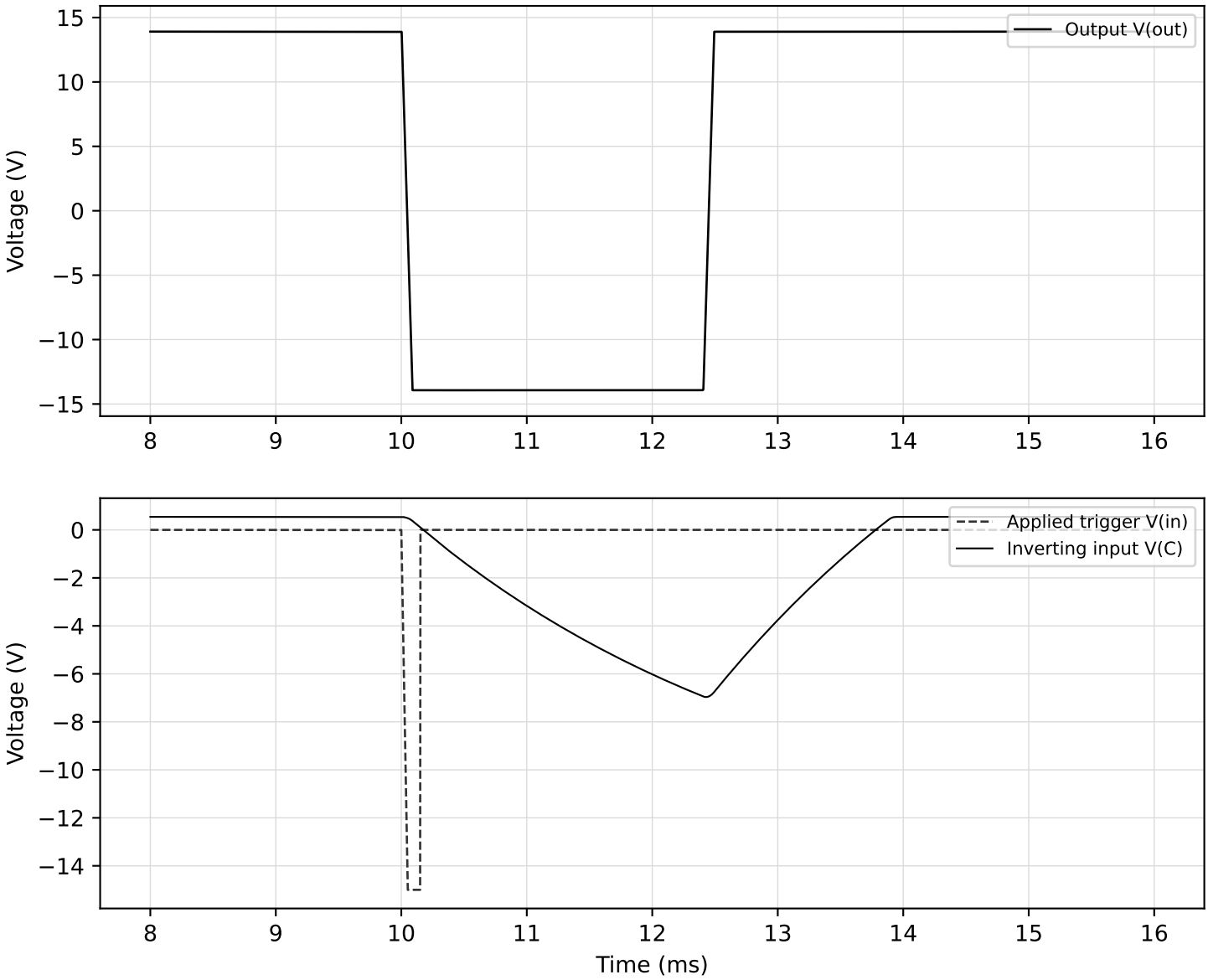
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EC3091D - Experiment 8

Monostable - Trial 3 / Figure 3

R = 33000 ohm; C = 0.1 uF; supplies = +/-15 V

NATIONAL SEMICONDUCTOR LM741 MACROMODEL



LOW pulse = 2.4044 ms; trigger = 50 Hz (external)

Measured from ngspice numerical transient data; 0 V output crossings.

Data: simulation_data/monostable_03.txt

Observation / handwritten notes: _____

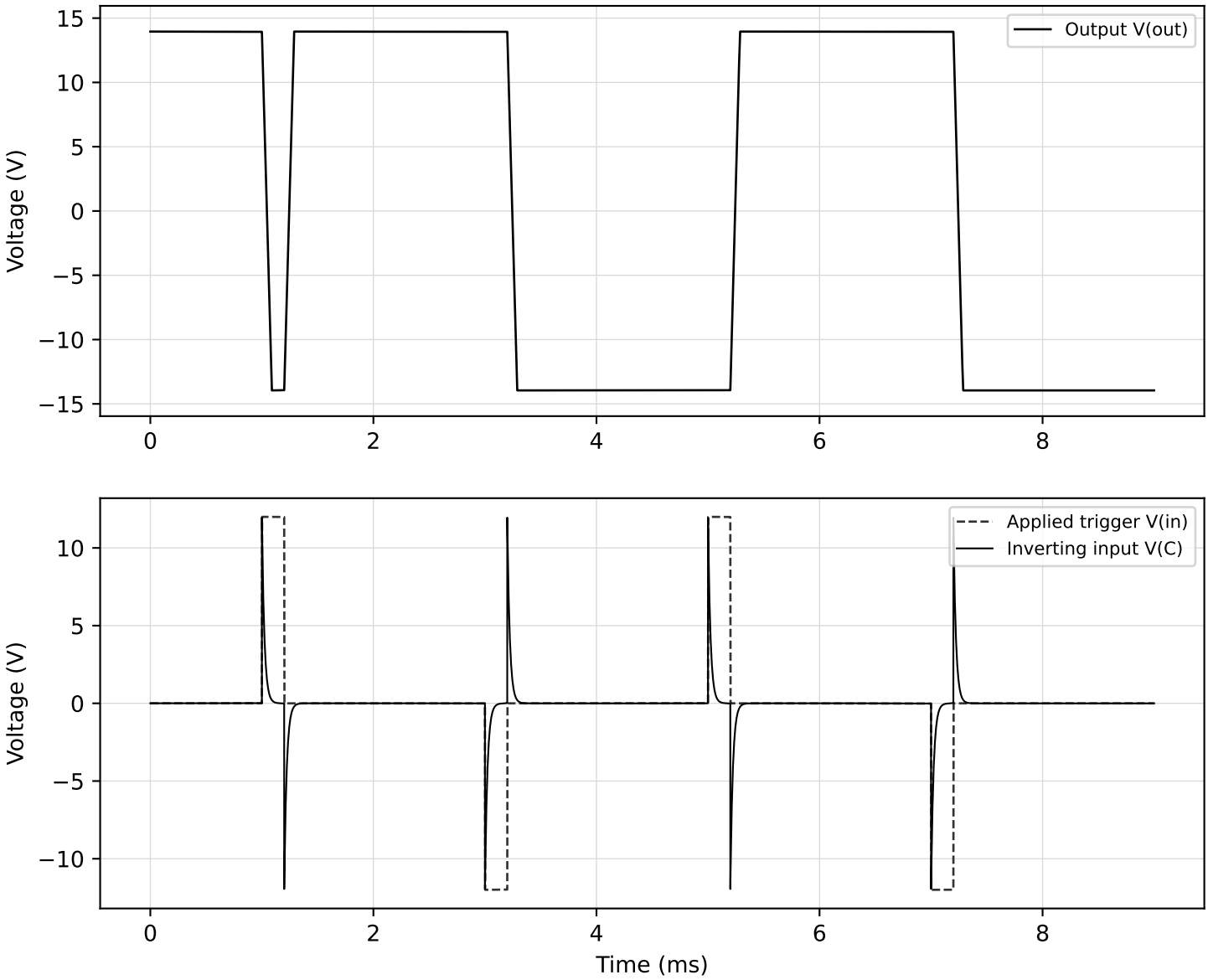
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EC3091D - Experiment 8

Bistable - Trial 2 / Figure 4

R = 220 ohm; C = 0.1 uF; supplies = +/-15 V

NATIONAL SEMICONDUCTOR LM741 MACROMODEL



Both states switch and persist; f, T, duty: N/A

Measured from ngspice numerical transient data; 0 V output crossings.

Data: simulation_data/bistable_02.txt

Observation / handwritten notes: _____

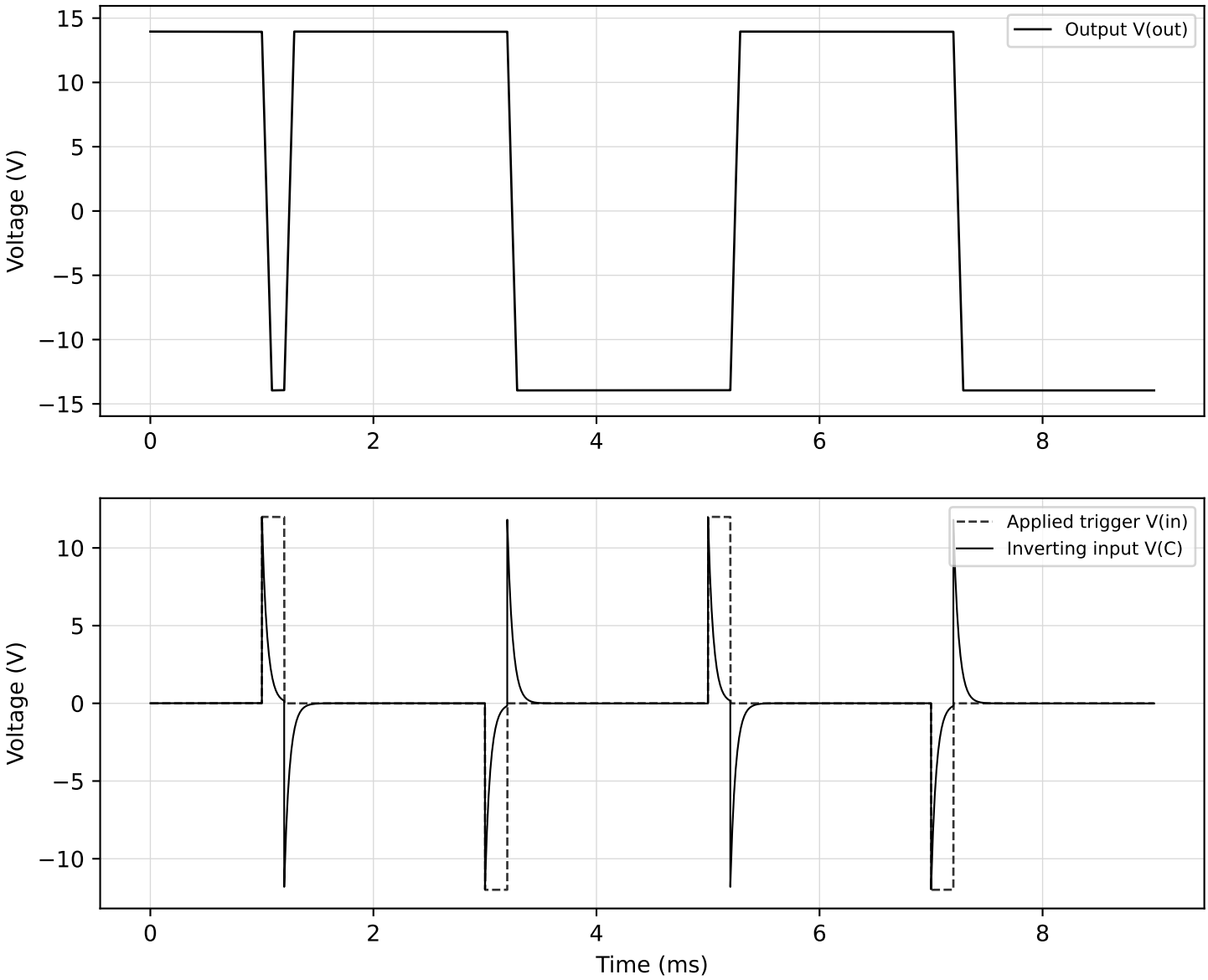
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EC3091D - Experiment 8

Bistable - Trial 3 / Figure 4

R = 470 ohm; C = 0.1 uF; supplies = +/-15 V

NATIONAL SEMICONDUCTOR LM741 MACROMODEL



Both states switch and persist; f, T, duty: N/A

Measured from ngspice numerical transient data; 0 V output crossings.

Data: simulation_data/bistable_03.txt

Observation / handwritten notes: _____

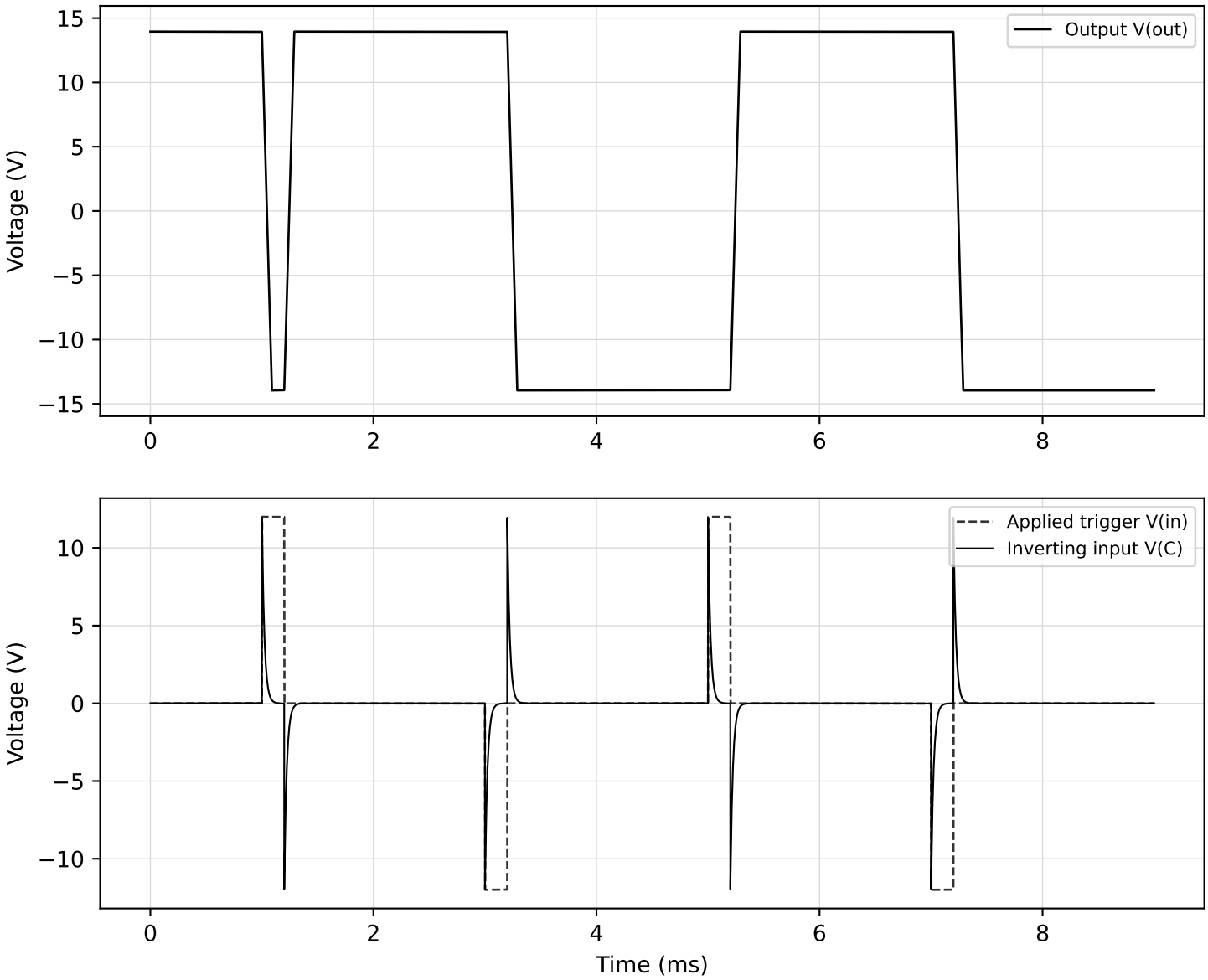
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EC3091D - Experiment 8

Bistable - Trial 4 / Figure 4

R = 100 ohm; C = 0.22 uF; supplies = +/-15 V

NATIONAL SEMICONDUCTOR LM741 MACROMODEL



Both states switch and persist; f, T, duty: N/A

Measured from ngspice numerical transient data; 0 V output crossings.

Data: simulation_data/bistable_04.txt

Observation / handwritten notes: _____

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